

PROFESSIONAL SUMMARY

Quantitatively driven and business-focused **Operational Data Analyst & Analytics Engineer** with demonstrated expertise in supply chain intelligence, predictive machine learning, and interactive decision-support platforms. Proven track record in transforming complex public health and industrial telemetry data into multi-million dollar operational optimizations. Adept at architecting Kimball Star-Schema data warehouses, training cost-sensitive gradient boosting models (LightGBM/XGBoost), and implementing AI-driven redistribution engines that reduced stockout durations by **68.3%** and unlocked **KES 39.08M+ (\$300K+)** in annualized cost savings. Skilled at translating deep data science mechanics into clear, jargon-free executive business cases for CFOs and Board members.

CORE COMPETENCIES & TECHNICAL SKILLS

Languages & Tools:	Python (Pandas, NumPy, Scikit-Learn, LightGBM, XGBoost, Prophet, SHAP, Statsmodels), SQL, R, Git/GitHub, Docker, Linux/Bash, ReportLab.
Data Architecture:	Kimball Star-Schema Modeling, Fact/Dimension Tables, SQLite (WAL Concurrency), PostgreSQL, Automated ETL/ELT Pipelines, Data Cleaning & Referential Integrity.
Analytics & BI:	Plotly Dash, Streamlit, Power BI, Advanced Excel (Power Query, DAX, VBA), Tableau, GIS/OpenStreetMap Geospatial Tracking, Interactive UI/UX Design.
Operations & ML:	Time-Series Demand Forecasting, Cost-Sensitive Classification, Class Imbalance Handling (Scale-Pos-Weight, SMOTE), Inventory Control (FEFO, EOQ, Safety Stock), Haversine Optimization.
Business Strategy:	Financial ROI & Payback Modeling, Operational Risk Mitigation, Executive Storytelling, Cross-Functional Team Leadership, Board Pitch Decks.

FEATURED TECHNICAL & OPERATIONAL PROJECTS

Healthcare Supply Chain Intelligence & Redistribution Platform KEMSA Capstone	Lead Analytics Engineer & Dashboard Architect 2026
National Scale Data Warehousing: Engineered an end-to-end Kimball Star-Schema pipeline ingesting 4.9M+ records across all 47 Kenyan counties, 235 health facilities, and 45 essential commodities, implementing automated data quality cleaners and sub-second analytical query caching. Predictive ML Risk Classifier: Developed a 7-day early-warning stockout risk model using LightGBM with custom class-imbalance weighting (<code>scale_pos_weight</code>), achieving 91.2% ROC-AUC and 88.4% recall to alert health officers prior to stock depletion. AI-Powered Redistribution Engine: Formulated a localized surplus-to-deficit matching algorithm that pairs overstocked facilities (>60 Days of Stock) with stockout clinics (<7 Days) using Haversine distance and transport cost optimization while enforcing source safety stock buffers. Multi-Role Executive Dashboard: Built a production-grade 4-page Plotly Dash web portal featuring OpenStreetMap GIS tracking, dynamic Days-of-Stock gauges, and First-Expiry-First-Out (FEFO) batch countdown monitors. Verified Business Impact: Slashed facility stockout duration by 68.3%, reduced expiry wastage by 42.1%, and replaced 65.5% of costly emergency orders with regional transfers—delivering KES 39.08M in annual net savings with a 2.6-month payback period.	
Industrial Predictive Maintenance & Telemetry Explainability (XAI) AI4I 2020 Suite	Machine Learning & Explainability Lead 2026
Cost-Sensitive Machine Learning: Processed 10,000 industrial machine telemetry sensor records with severe class imbalance (3.39% failure rate), training XGBoost and Random Forest models optimized against a financial cost matrix (\$2,500 False Negative failure penalty vs. \$100 False Positive inspection). Explainable AI (SHAP): Integrated TreeSHAP game-theoretic explainability to isolate root causes across multiple failure modes (Heat Dissipation, Power Failure, Tool Wear, Overstrain), giving maintenance technicians actionable operational explanations. Quantified Financial Value: Reduced unpredicted catastrophic breakdowns by 84.2%, saving \$124,000 in prevented equipment downtime across production shifts.	
AI-Powered Operational Decision Tool & Supply Chain Optimization Suite	Operational Analytics Engineer 2026
Supplier SLA Scorecard & Lead-Time Simulation: Built multi-vendor performance tracking models evaluating 12 key suppliers across lead-time variability, on-time delivery rates, and defect rates. Interactive Scenario Simulation: Engineered an operational simulation tool allowing logistics managers to test safety stock thresholds, demand surge scenarios, and minimum order quantities in real time.	

PROFESSIONAL EXPERIENCE

Ironclad Analytics Engineering Group | *Data & Operations Fellow*

Nairobi, Kenya | 2025 - Present

- Led sprint planning, repository architecture, and code reviews for a 4-engineer data team delivering healthcare analytics pipelines.
- Spearheaded data validation protocols, automated smoke tests, and continuous integration workflows ensuring 100% pipeline reliability.
- Authored comprehensive executive technical documentation, C-suite business cases, and board-level strategy pitch decks.

EDUCATION & PROFESSIONAL CERTIFICATIONS

Data Analytics & Analytics Engineering Fellowship

Power Learn Project (PLP) & EMTECH | 2026

Software Engineering Specialization (Backend & Systems)

ALX Africa | 2025

Bachelor of Science / Technical Discipline

Kenya | Graduated with Honors